11 The Metapopulation Room

Patch occupancy

CORE PATH · BSC	EXTENSION · MSC	BEFORE THIS ROOM
about 34 min	**+ 28 min**	The Crowding Room (02), whose equilibrium this room's p* is compared with.
steps 1–4	steps 5–7	

OPEN THE ROOM

`https://evolinus.github.io/zooe/teaching/population-laboratories/` — then the Metapopulation tab

Every other room counts individuals; this one counts populations. Each patch is occupied or empty, occupied patches go extinct, and empty ones are colonised from occupied ones. This guide asks how the whole can persist when every part dies, and what habitat loss does to it.

LEARNING OBJECTIVES

After the core path you should be able to

- Explain how a metapopulation persists through a balance of colonisation and extinction.
- Predict the equilibrium occupancy $p^* = 1 - e/c$ and check it.
- Show that a steady occupancy hides continual turnover of individual patches.
- Explain how destroying habitat lowers p^*, and why extinction can happen with habitat still standing.

The extension adds

- Show that only the ratio e/c sets p^*, while the rates set the turnover.
- Explain why small networks are at risk even when $p^* > 0$.
- Calculate the extinction threshold for habitat loss, $h = 1 - e/c$.

SUGGESTED ROUTES

- **35-minute practical (BSc):** steps 1–4 and questions 1–3.
- **Landscape conservation session:** steps 4 and 7 with question 3.
- **MSc:** steps 5–7 and questions 4–6 (about 30 minutes).

How to work through it. Every step asks you to *predict* before you press anything, and to *explain* afterwards. Write the prediction down and commit to it: a wrong prediction you can account for teaches more than a right one you guessed. Runs are random, so where a step says so, run it more than once before you decide what happened.

1 Occupancy settles

10 min

SET UP

The defaults: 100 patches, p_0 = 0.10, c = 0.50, e = 0.20, time span 80, 1 replicate.

PREDICT FIRST

Calculate $p^* = 1 - e/c$. Will the landscape settle there?

RUN

Press **Run**.

LOOK AT

the solid line against the dashed equation and the purple p^* line, and the *Colonisation and extinction vs. occupancy* panel.

EXPLAIN

Why is colonisation a hump and extinction a straight line? Where do they cross?

2 Nothing steady underneath

8 min

SET UP

The same run.

PREDICT FIRST

Once occupancy has flattened, will the grid stop changing?

RUN

Park the History Scrubber after the curve flattens and step forward one frame at a time. Watch the two turnover counters.

EXPLAIN

“Persistence at the landscape scale is not the absence of extinction — it is extinction being outrun.” Explain with what you saw.

3 Extinction faster than colonisation

8 min

SET UP

e = **0.60**, c = 0.50.

PREDICT FIRST

Every patch is just as good as before. What will happen to the metapopulation?

RUN

Run, with 5 replicates.

EXPLAIN

Nothing is wrong with any individual patch. Why does the whole thing still empty?

4 Destroying habitat

8 min

SET UP

Back to e = 0.20. Open *More parameters*: habitat destroyed h = 0.20, then 0.40, then 0.60.

PREDICT FIRST

Calculate $p^* = 1 - h - e/c$ for each. At which h will the species disappear?

RUN

Run each.

EXPLAIN

The species occupied only 60% of the landscape before any habitat was lost. Why, then, is destroying 60% of the patches fatal, with patches still standing empty?

EXTENSION · MSC

5 Only the ratio

10 min

SET UP

h = 0. c = 0.50, e = 0.20; then c = **1.00**, e = **0.40**.

PREDICT FIRST

Will p^* change? Will the turnover counters?

RUN

Run both and compare the counters at the end.

EXPLAIN

What is it that the ratio fixes, and what do the rates themselves fix?

6 Small networks

10 min

SET UP

c = 0.50, e = **0.40** (p^* = 0.2), 20 replicates. Number of patches H = **16**, then H = **400**.

PREDICT FIRST

In which network will replicates go extinct, although p^* is the same?

RUN

Run each.

EXPLAIN

Why does the equation's prediction mislead for small networks?

7 The threshold

8 min

SET UP

c = 0.50, e = 0.20.

PREDICT FIRST

Calculate the largest h the species can survive. If 10% of patches are lost each decade, how many decades until the metapopulation fails?

RUN

Check with h just below and just above your threshold.

EXPLAIN

The room destroys habitat before the run starts. What would an *extinction debt* look like if habitat were lost gradually during the run?

DISCUSSION AND WORKSHEET QUESTIONS

1. BSc The Glanville fritillary butterfly in Åland lives in hundreds of meadows, most of which lose their population every few years. Explain how the species persists.
2. BSc Why is it possible for a species to vanish from a landscape where plenty of suitable habitat remains?
3. BSc Compare p^* with the carrying capacity K of the Crowding Room. What is similar, and what is different?
4. MSc The rescue effect makes extinction less likely in a well-occupied landscape. How would it change the shape of the extinction line and the outcome?
5. MSc Levins' patches have no positions. How would distance-limited colonisation change the room's conclusions?
6. MSc Explain why protecting only currently occupied patches might fail to protect a metapopulation.

CHECKING YOUR REASONING

Read this after you have written your own answers. These are what the model should show on most runs — not the only acceptable explanations.

- **Step 1.** $p^* = 1 - 0.2/0.5 = 0.6$, and the run settles there. Colonisation needs both occupied senders and empty receivers, so it is a hump; extinction is a constant risk per occupied patch, so a straight line.
- **Step 2.** The grid keeps changing every frame and both counters keep climbing, while the fraction barely moves.
- **Step 3.** With $e > c$ the landscape empties: there is no occupancy at which colonisation outpaces extinction.
- **Step 4.** $p^* = 0.4$, 0.2 and 0. At $h = 0.6$ the colonisation term can no longer keep up, and the species is lost although 40% of the patches are still habitat.
- **Step 5.** p^* stays at 0.6; turnover doubles. The ratio fixes the equilibrium; the rates fix how fast patches change hands.
- **Step 6.** With 16 patches, $p^* = 0.2$ means about three occupied patches, and chance often takes them all; with 400 the network persists.
- **Step 7.** $h = 1 - e/c = 0.6$. Losing 10% a decade, the network fails after about six decades — and with gradual loss, occupancy lags above what the landscape can support, so the decline continues after the loss stops.